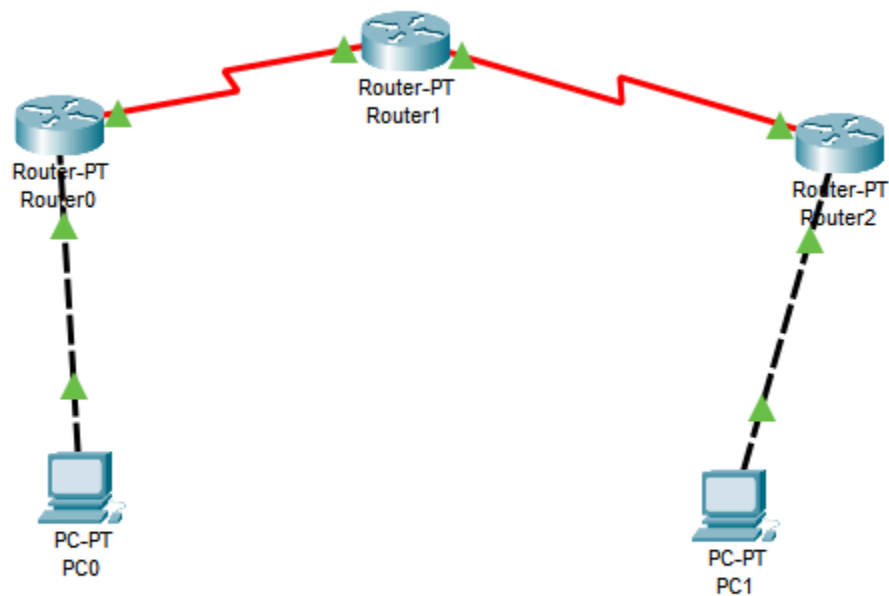


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Experiment: Configure IP address to routers
in packet tracer. Explore the following
messages: ping responses, destination
unreachable, request timed out, reply

Topology Diagram:



Verify Routing Tables:

1. R1
show ip route

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, Serial2/0
S    40.0.0.0/8 [1/0] via 20.0.0.2

Router#

```

2. R2
show ip route

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is directly connected, Serial2/0
C    30.0.0.0/8 is directly connected, Serial3/0
S    40.0.0.0/8 [1/0] via 30.0.0.2

```

3. R3
show ip route

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
C    30.0.0.0/8 is directly connected, Serial2/0
C    40.0.0.0/8 is directly connected, FastEthernet0/0

Router#

```

Test Connectivity Step-by-Step:

1. Test 1: PC0 → R1
On PC0:
ping 10.0.0.1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2. Test 2: R1 → R2

On R1:

ping 20.0.0.2

```
Router#ping 20.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/11/16 ms
```

3. Test 3: R2 → R3

On R2:

ping 30.0.0.2

```
Router#ping 30.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/13/18 ms

Router#
```

4. Test 4: R3 → PC1

On R3:

ping 40.0.0.10

```
Router#ping 40.0.0.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 40.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Final PC0 → PC1 Test

Now go to:

PC0 → Desktop → Command Prompt

Enter:

ping 40.0.0.10

```
C:\>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=9ms TTL=125
Reply from 40.0.0.10: bytes=32 time=21ms TTL=125
Reply from 40.0.0.10: bytes=32 time=22ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 22ms, Average = 13ms

C:\>
```

Finally: Demonstrate the 3 Ping

Responses

Once the normal configuration is working, your experiment asks you to explore three results.

1. Reply

Normal configuration:

PC0> ping 40.0.0.10

```

C:\>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=17ms TTL=125
Reply from 40.0.0.10: bytes=32 time=23ms TTL=125
Reply from 40.0.0.10: bytes=32 time=20ms TTL=125
Reply from 40.0.0.10: bytes=32 time=13ms TTL=125

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 23ms, Average = 18ms

C:\>|

```

2. Request Timed Out

Temporarily disable R3's connection to PC1:

R3# configure terminal

R3(config)# interface fastEthernet 0/0

R3(config-if)# shutdown

Then:

PC0> ping 40.0.0.10

```

C:\>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Request timed out.
Reply from 40.0.0.10: bytes=32 time=21ms TTL=125
Reply from 40.0.0.10: bytes=32 time=23ms TTL=125
Reply from 40.0.0.10: bytes=32 time=23ms TTL=125

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 23ms, Average = 22ms

```

3. Destination Unreachable

Temporarily remove R1's route:

R1# configure terminal

R1(config)# no ip route 40.0.0.0 255.0.0.0 20.0.0.2

Then:

PC0> ping 40.0.0.10

```
C:\>ping 40.0.0.10
```

```
Pinging 40.0.0.10 with 32 bytes of data:
```

```
Reply from 30.0.0.2: Destination host unreachable.
```

```
Reply from 30.0.0.2: Destination host unreachable.
```

```
Reply from 30.0.0.2: Destination host unreachable.
```

```
Reply from 30.0.0.2: Destination host unreachable.
```

```
Ping statistics for 40.0.0.10:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>
```